CASE STUDY 1 · SHARE PHASE

How members and casual riders use Cyclistic bikes differently

Cyclistic Bike-Share · Analysis of 2019 trip data

Presented by

Fizaan Ali

Junior Data Analyst

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Prepared for

Lily Moreno and the executive team

OUTLINE

What we will cover

01 **Business task**
The question and who it serves

02 **Analysis**
Data sources, cleaning and approach

03 **Problem**
How members and casual riders differ

04 **Aha moment**
Casual riders carry over half of ride time

05 **Conclusion**
Findings, recommendations, next steps

06 **Appendix**
Data dictionary, tables and limitations

SECTION

01

Business task

What we were asked to find out, and who needs the answer.

BUSINESS TASK

The task in one sentence

Analyse how casual riders and annual members use Cyclistic bikes differently, so marketing can design strategies that convert casual riders into annual members.

GUIDING QUESTION

How do annual members and casual riders use Cyclistic bikes differently?

WHY IT MATTERS

Finance found annual members much more profitable. Marketing believes growth depends on adding members.

BUSINESS TASK

Key stakeholders

Lily Moreno

Director of Marketing

WHAT THEY NEED

Clear behaviour differences to build campaigns on.

Analytics team

Marketing analysts

WHAT THEY NEED

Reliable, documented analysis to build on.

Executive team

Final decision makers

WHAT THEY NEED

Evidence and clear visuals to approve the plan.

SECTION

02

Analysis

Where the data came from, how it was cleaned and how we measured.

ANALYSIS

Data sources

3,817,991 trips across all four quarters of 2019, after cleaning

Q1	Q2	Q3	Q4
Jan to Mar	Apr to Jun	Jul to Sep	Oct to Dec
Divvy_Trips_2019_Q1.csv	**Divvy_Trips_2019_Q2.csv**	**Divvy_Trips_2019_Q3.csv**	**Divvy_Trips_2019_Q4.csv**

Source and licence

Divvy trip data from Motivate International, used under the Divvy Data License Agreement.

Privacy

No personally identifiable information used. Gender and birth year were dropped.

Credibility

Recorded by the system itself. Limit: 2019 only, and rides are counted, not people.

ANALYSIS

Cleaning process

1 Combine files

Four quarterly files merged. Q2 headers renamed to match the others.

2 Drop extra fields

Removed gender and birthyear: not needed and partly missing.

3 Fix data types

Timestamps parsed as dates. Commas removed from tripduration.

4 Relabel customers

Subscriber became Member. Customer became Casual.

5 Create new fields

ride_length in seconds and day_of_week from start_time.

6 Remove bad rows

Dropped 13 rides with zero or negative length.

Result: 3,817,991 trips ready for analysis. Rides over 24 hours were reviewed but kept.

ANALYSIS

How we measured

1

Who rides

Ride counts by customer type.

2

When they ride

Day of week, and weekday versus weekend share.

3

How long they ride

Mean and median ride length in minutes.

Tools: Python with pandas and matplotlib in Jupyter Notebook. The mean shows total riding effort; the median shows a typical ride.

SECTION

03

Problem

Members and casual riders use the same bikes on different days and for different lengths of time.

PROBLEM

Who rides: members vs. casual riders



Member rides

2,937,360

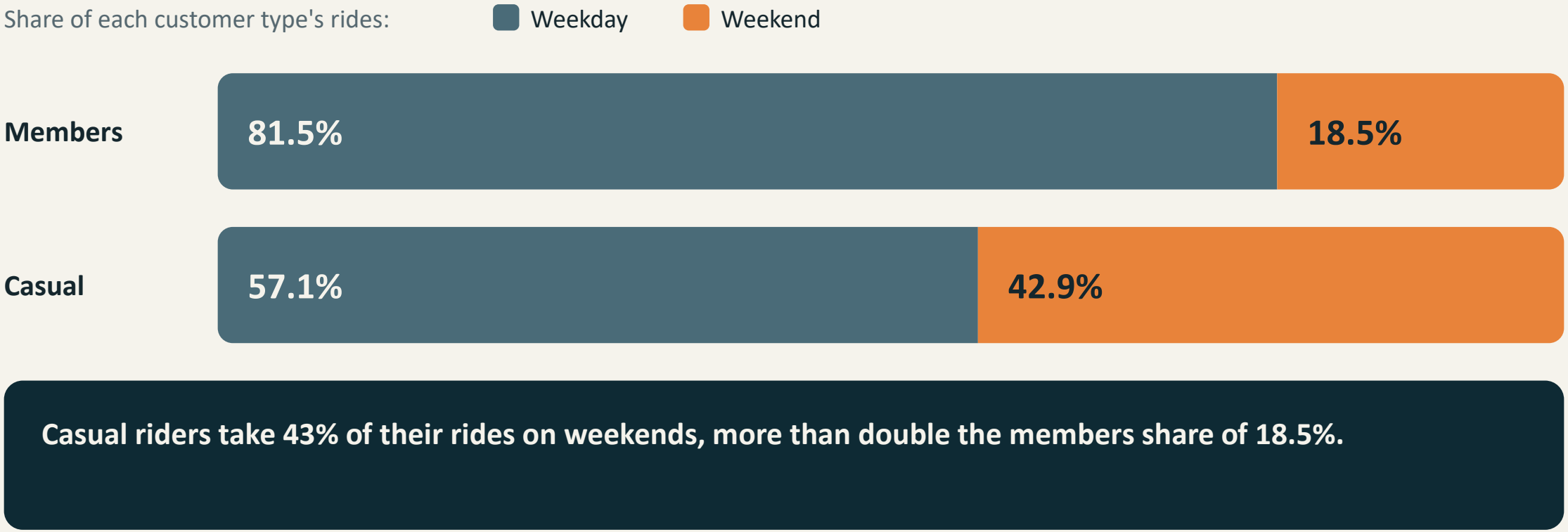
Casual rides

880,631

About three of every four rides already come from members. Casual riders are the smaller group.

PROBLEM

Weekday and weekend riding

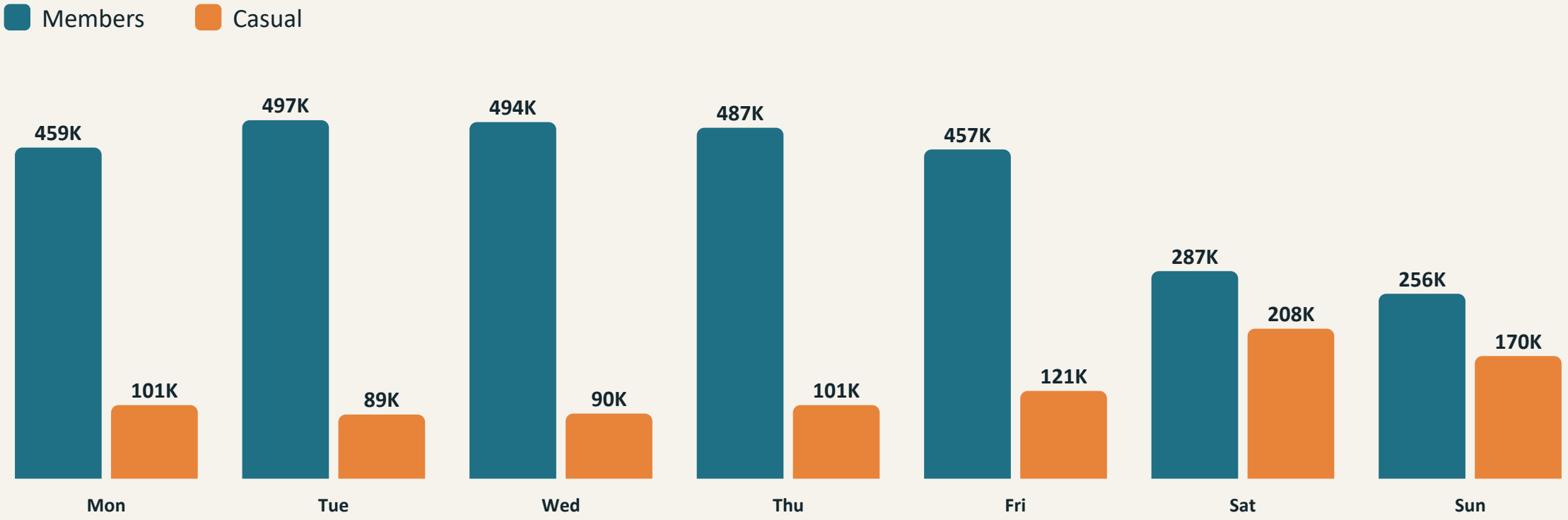


Casual riders take 43% of their rides on weekends, more than double the members share of 18.5%.

PROBLEM

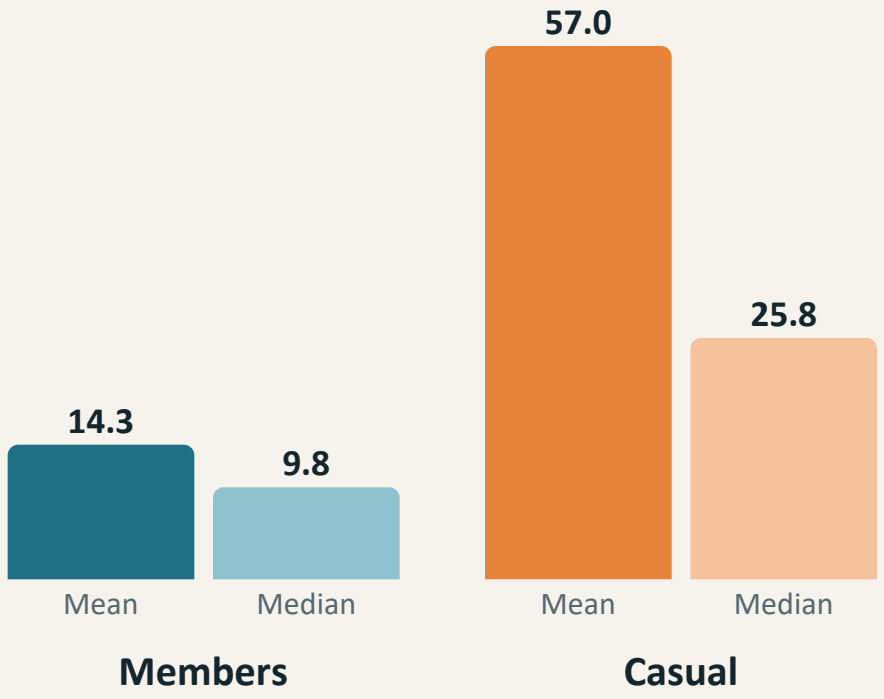
Rides by day of week

Members peak on Tuesday. Casual riders peak on Saturday.



PROBLEM

Ride length (minutes)



4.0×

longer rides on average: 57.0 vs. 14.3 minutes

2.6×

longer at the median: 25.8 vs. 9.8 minutes

The mean is pulled up by a few very long rides. The median shows the typical ride.

SECTION

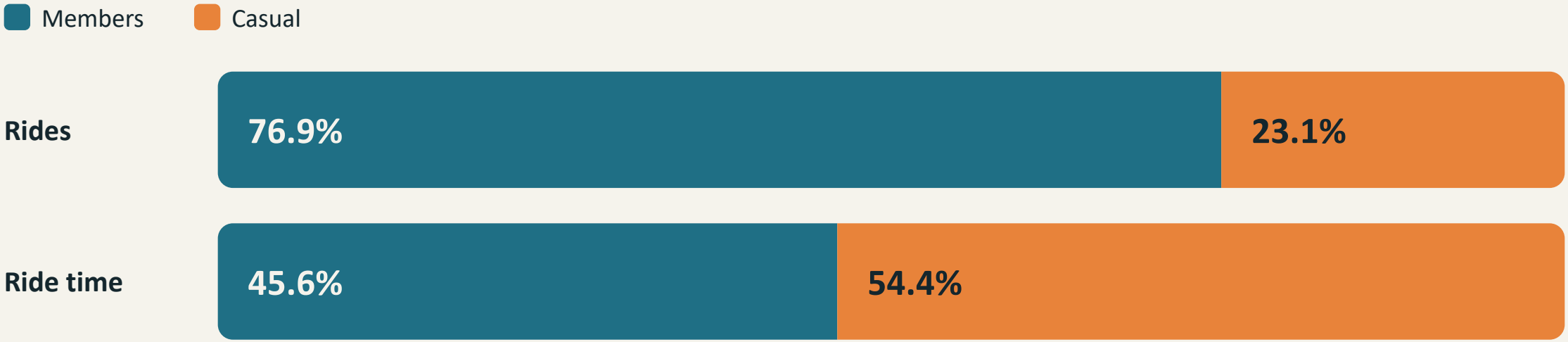
04

Aha moment

Casual riders make a quarter of the rides but carry over half of the riding time.

AHA MOMENT

Share of rides vs. share of ride time



Casual riders are 23% of rides but about 54% of all time spent riding. They are worth more attention than their ride count suggests.

Ride time = rides × mean ride length. Includes very long rides, which were kept in the data.

AHA MOMENT

Two riding profiles

Members: the weekday rider

81.5%	of rides on weekdays
9.8 min	median ride length
Tuesday	busiest day, 497K rides
2.94M	rides in 2019

Casual: the weekend rider

42.9%	of rides on weekends
25.8 min	median ride length
Saturday	busiest day, 208K rides
881K	rides in 2019

Trip purpose is not recorded. Commuting and leisure labels are our reading of the pattern.

SECTION

05

Conclusion

What we found, what we recommend and what to do next.

CONCLUSION

Key findings

FINDING 1

43% vs. 18%

Different calendars

Share of rides on weekends, casual vs. members.

FINDING 2

26 vs. 10 min

Different ride lengths

Median ride, casual vs. members.

FINDING 3

54%

Hidden weight

Share of all riding time that comes from casual riders.

CONCLUSION

Top three recommendations

1

Offer a weekend membership

Give casual riders a weekend-focused membership or trial. Their riding peaks on Saturday and Sunday.

Based on: 43% of casual rides at weekends

2

Sell value on longer rides

Build the message around longer trips and show how membership can lower the cost of frequent riding.

Based on: median casual ride of 25.8 minutes

3

Time digital prompts

Place membership offers in email, social and the app on Fridays and weekends, when casual riding climbs.

Based on: casual rides rise from Friday to Saturday

CONCLUSION

Next steps and further data

NEXT STEPS

- Check plan prices against casual ride lengths
- Pilot a weekend offer and measure sign-ups
- Move on to why casual riders would buy, and digital media

FURTHER DATA TO EXPLORE

- Time of day and month-by-month seasonality
- Start and end stations, already in the data
- Pass type and pricing data
- More recent trip data

SECTION

06

Appendix

Reference material: fields used, supporting numbers and limitations.

APPENDIX

A1 · Data dictionary

Field	Description
trip_id	Unique ID for each ride
start_time, end_time	Ride start and end, parsed as dates
tripduration	Supplied duration in seconds (commas removed)
from_station_name, to_station_name	Start and end stations (not analysed yet)
usertype	Member (was Subscriber) or Casual (was Customer)
ride_length	end_time minus start_time, in seconds
day_of_week	0 = Monday to 6 = Sunday, from start_time
day_type	Weekday (0 to 4) or Weekend (5 and 6)

APPENDIX

A2 · Rides by day of week

Day	Member rides	Casual rides	Casual share of day
Monday	458,780	101,489	18.1%
Tuesday	497,025	88,655	15.1%
Wednesday	494,277	89,745	15.4%
Thursday	486,915	101,372	17.2%
Friday	456,966	121,141	21.0%
Saturday	287,163	208,056	42.0%
Sunday	256,234	170,173	39.9%
Total	2,937,360	880,631	23.1%

APPENDIX

A3 · Limits and good practice

LIMITS OF THIS ANALYSIS

- 2019 data only; behaviour may have changed
- Rides are counted, not individual people
- Mean ride length is lifted by very long rides, which were kept
- Reason for riding is not recorded
- Day of week is based on ride start time

GOOD PRACTICE FOLLOWED

- No personally identifiable information used
- Data used under the Divvy licence
- Every cleaning step documented in the notebook
- Findings reported with both mean and median